Problem 6: Campus Portal Browser History

TASK

A campus portal browser opens on a home page and supports three actions: visit a new page (which clears all forward history), go back up to x pages, and go forward up to x pages. Back and forward stop at the first and last pages available.

Design a `BrowserHistory` class backed by your own doubly linked list, with `visit(url)`, `back(steps)` and `forward(steps)`; back and forward return the current page. Use your own node class (for example `class Node { int data; Node prev, next; }`); do not use `java.util.LinkedList` or other collections for the list itself.

CONSTRAINTS

- 1 <= number of actions <= 5000
- 1 <= steps <= 100
- Each visit is O(1); back and forward are O(steps)

EXAMPLE

```
home = srm.edu
visit a.com, visit b.com, visit c.com
back 1, back 1, forward 1, visit d.com, forward 2, back 2, back 7

Expected Output: b.com, a.com, b.com, d.com, a.com, srm.edu
```

COMPLEXITY REQUIREMENTS

State the cost of each operation and the space used. Explain how a visit discards the forward history and what happens to those nodes.

HINT

> *Keep a reference to the current node; visit links a new node after it and drops everything that followed.*
